

Effects of Açai (*Euterpe oleracea* Mart.) berry preparation on metabolic parameters in a healthy overweight population: A pilot study

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Background The purpose of this study was to evaluate the effect of açai fruit pulp on risk factors for metabolic disorders in overweight subjects. The açai palm (*Euterpe oleracea* Mart.), which is native to South America, produces a small, black-purple fruit which is edible. The fruit has recently become popular as a functional food due to its antioxidant potential. Although several studies have been conducted in vitro and with animals, little is known about the potential health benefits in humans aside from an increase in plasma anti-oxidant capacity. Metabolic syndrome is a condition which is defined by a cluster of risk factors for cardiovascular disease and/or type-2 diabetes. Preliminary studies indicate that a reduction in reactive oxygen species can assist in the normalization of the metabolic pathways involved in this syndrome. **Methods** This was an open label pilot study conducted with 10 overweight adults ($BMI \geq 25 \text{ kg/m}^2$ and $\leq 30 \text{ kg/m}^2$) who took 100 g açai pulp twice daily for 1 month. The study endpoints included levels of fasting plasma glucose, insulin, cholesterol, triglycerides, exhaled (breath) nitric oxide metabolites (eNO) and plasma levels of high sensitivity C-reactive protein (hs-CRP). The response of blood glucose, blood pressure and eNO to a standardized meal was determined at baseline and following the 30 day treatment. **Results** Compared to baseline, there were reductions in fasting glucose and insulin levels following the 30 day treatment (both $p < 0.02$). There was also a reduction in total cholesterol ($p = 0.03$), as well as borderline significant reductions in LDL-cholesterol and the ratio of total cholesterol to HDL-cholesterol (both $p = 0.051$). Compared to baseline, treatment with açai ameliorated the post-prandial increase in plasma glucose following the standardized meal, measured as the area under the curve ($p = 0.047$). There was no effect on blood pressure, hs-CRP or eNO. **Conclusion** In this uncontrolled pilot study, consumption of açai fruit pulp reduced levels of selected markers of metabolic disease risk in overweight adults, indicating that further studies are warranted.